Obaidullah Miakhil

Machine Learning / AI Engineer

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Islamabad, Pakistan · Open to remote and relocation

ML engineer with hands-on experience taking deep learning models from raw data to evaluated, deployable pipelines, mainly in healthcare. I build computer vision and sequence models in PyTorch, add uncertainty estimation and explainability so models know when to defer to a human, and run rigorous, leakage-free evaluation on limited compute. MS in AI (CGPA 4.00/4.00), one Q1 journal publication and three first-author papers under review.

SKILLS

ML / DL	PyTorch, TensorFlow, Keras, scikit-learn · CNNs (EfficientNet), RNN/LSTM, attention, multi-label and weakly supervised learning, time-series forecasting
Trustworthy ML	Uncertainty estimation (MC Dropout), calibration, selective prediction, Grad-CAM++ and SHAP explainability, leakage-safe evaluation, confidence intervals
Data and imaging	DICOM/CT processing, HU windowing, large-scale data pipelines, embedding caching, tabular healthcare data, Power BI dashboards
GenAI and apps	LLM chatbots, Hugging Face, Flask APIs, ASP.NET, SQL Server
Programming	Python, SQL, C++, C#, MATLAB · Git/GitHub, Kaggle GPU workflows, Jupyter

EXPERIENCE

Research Engineer — Sino-Pak Center for Artificial Intelligence (SPCAI), PAF-IAST — Jul 2025 – Present

- Built an end-to-end deep learning pipeline for intracranial hemorrhage detection on head CT: DICOM ingestion, HU windowing and patient-level splits that prevent data leakage.
- Designed a sequence-aware model (EfficientNet slice encoder + bidirectional LSTM, focal loss) for multi-label classification of five hemorrhage subtypes.
- Added Monte Carlo Dropout uncertainty and a selective-refusal rule that routes uncertain scans to a radiologist, with Grad-CAM++ heatmaps for localization.
- Cut compute needs by caching slice embeddings, so training and repeated uncertainty inference run on two consumer GPUs.

AI Research Intern (Remote) — Cube Healthcare, Michigan, USA — Jun – Aug 2024

- Built an insurance claim denial prediction model reaching 93% accuracy on real payer data, plus a rule-based pre-submission claim scrubber.
- Integrated the models with claim management workflows; the work was later published in *Frontiers in Health Services* (Q1, 2026).

Apprentice, LLM Chatbot — SPCAI, PAF-IAST — Jun – Aug 2025

- Developed an LLM-based chatbot that answers prospective students' questions and guides them through university admission.

Generative AI Intern — Insafdaar, Abbottabad — Jul – Sep 2024

- Built generative AI solutions for legal-domain problems as part of a team of AI interns.

Lab Engineer and Professional BI Trainer — PAF-IAST; Khyber Pakhtunkhwa IT Board (KPITB) — Jul 2025 – Present

- Teach the Object-Oriented Programming lab to undergraduates; delivered a 144-hour Power BI and business intelligence program to 50+ graduates for KPITB (Jun – Sep 2026).

Software Development Internships — ByteEdge Software Solutions; PAF-IAST Software Cell — 2022 – 2024

- Built a hospital management system in ASP.NET, Windows LMS features on SQL Server, and customized the university's Moodle CMS.

SELECTED PROJECTS

- **Weakly supervised hemorrhage localization (MS thesis):** localizes bleeding on CT and classifies five subtypes from scan-level labels only, using class activation maps and attention on RSNA and PhysioNet BHX data.
- **Fair benchmarking of deep learning forecasters:** evaluated deep learning return-forecasting models across eight equity indices under a symmetric benchmark design.

EDUCATION

M.S. Artificial Intelligence — 2025 – Feb 2027 (exp.)
PAF-IAST, Pakistan · CGPA 4.00/4.00

B.S. Computer Science — 2021 – 2025
PAF-IAST, Pakistan · Allama Iqbal Scholarship (HEC)

PUBLICATIONS

- Jadoon, M. M., **Miakhil, O.**, et al. (2026). A machine learning framework for predicting healthcare insurance claim denials. *Frontiers in Health Services*, 6. Q1. DOI
- Three first-author papers under review (CT hemorrhage detection, uncertainty and selective refusal; financial forecasting benchmarks).

CERTIFICATIONS AND LANGUAGES

Google Advanced Data Analytics (2025) · ML Engineer Fundamentals, LinkedIn (2025) · Nuts and Bolts of ML, Google (2024) · English (C1), Urdu (C1), Pashto and Dari (native)